Greenhouse gas inventory 2025

HTWest (representative data), Bonn, Germany. Contract precision cutting and converting of filter media. Scope 1 and 2, GHG Protocol Corporate Standard.

Representative data. The figures in this edition are realistic estimates for a precision cutting site of this size, not HTWest's own meter readings. The method and factors are real; the activity numbers will be replaced with invoice data.

Summary

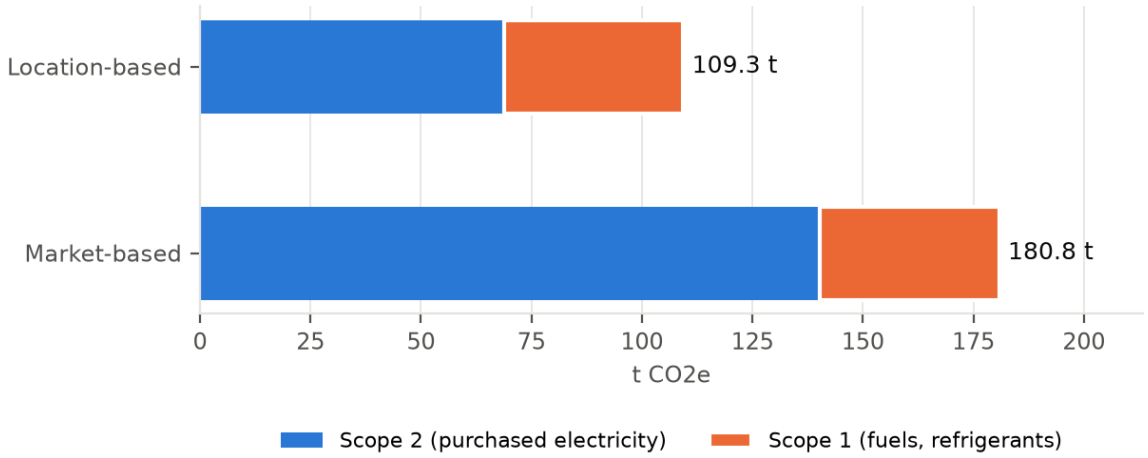
In 2025 the site emitted **109.3 t CO_2e** across Scope 1 and 2 on a location-based view, and **180.8 t** on a market-based view. The gap is all electricity: the grid average is one number, the contract the site buys under is another.

Purchased grid electricity is the largest source at 68.8 t, 63% of the location-based total. Natural gas boiler, space heating comes next at 32.9 t. Everything else together is 7%.

Market-based Scope 2 is higher than location-based. That is expected for a German company on a standard tariff: the renewable power on the grid has mostly been claimed by others through guarantees of origin, so unclaimed electricity carries the German residual mix, which is dirtier than the average. Customers who ask for a market-based figure will see the larger number.

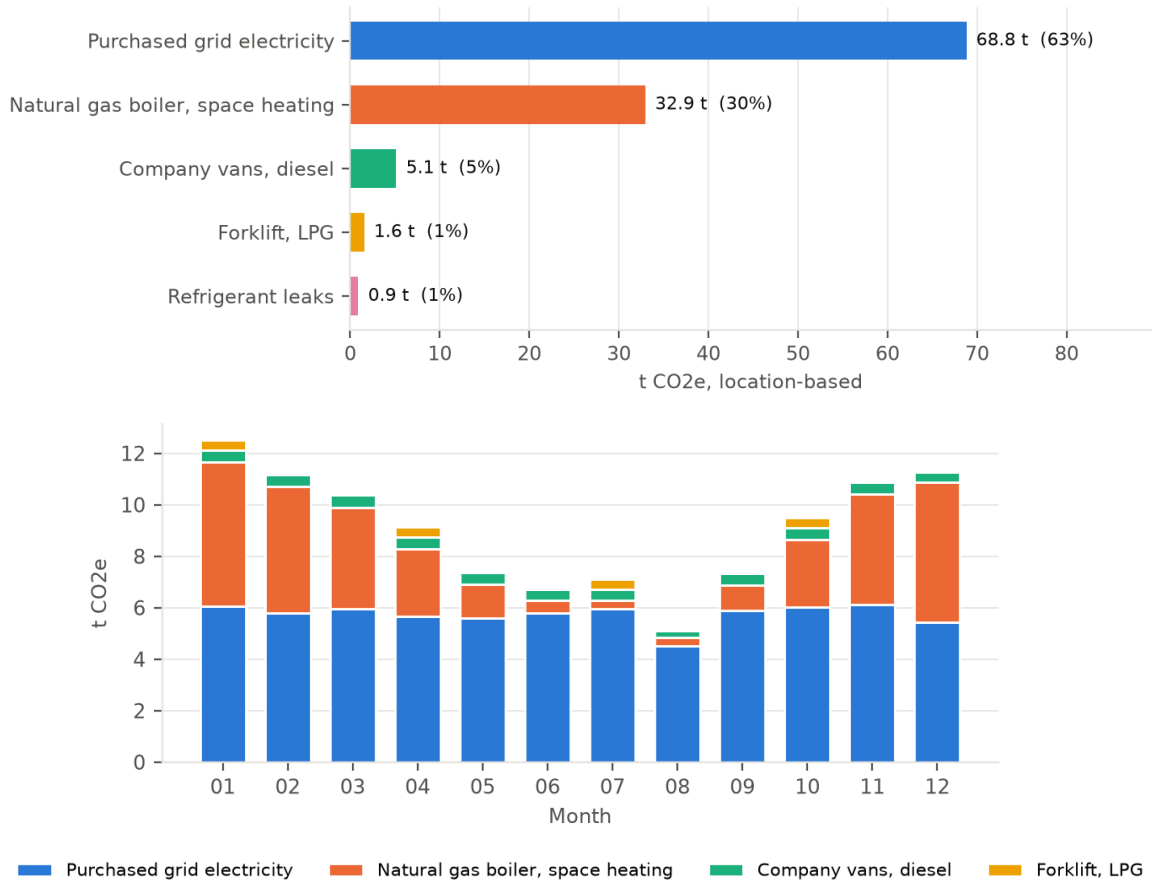
	t CO_2e	Share	95% range (t)
Scope 1	40.5	37%	39.5 to 41.6
Stationary combustion	32.9		
Mobile combustion	6.7		
Fugitive (refrigerants)	0.9		
Scope 2, location-based	68.8	63%	65.3 to 72.2
Scope 2, market-based	140.3		133.2 to 147.4
Total, location-based	**109.3**	100%	105.7 to 112.9
Total, market-based	**180.8**		173.7 to 188.0

Biogenic CO_2 from biodiesel, reported outside the scopes: 0.28 t.



Where the emissions come from

Electricity drives the cutting lines, compressors, dust extraction and HVAC, so it dominates. Gas heats the building and follows the weather: most of it burns between November and March. Refrigerant leaks add 0.9 t, small but not zero, and the easiest Scope 1 source to miss.



Monthly figures are location-based. Refrigerant leaks are annual and not shown by month.

Line items

Source	Scope	Quantity	Factor	DQ	t (LB)	t (MB)
Natural gas boiler, space heating	1	180,000 kWh	DESNZ 2025	1	32.9	32.9
Company vans, diesel	1	2,000 litre	DESNZ 2025	2	5.1	5.1
Forklift, LPG	1	1,000 litre	DESNZ 2025	2	1.6	1.6
Refrigerant leaks (all equipment)	1	0.51 kg released	IPCC AR6 GWPs	2-4	0.9	0.9
Purchased grid electricity	2	200,000 kWh	UBA 2025 (LB); AIB 2025 residual mix (MB)	1	68.8	140.3
Total					**109.3**	**180.8**

DQ is the data quality score (1 best, 5 worst). LB and MB are location- and market-based. Every line traces back to rows in the input workbook; see inventory_lines.csv.

Energy and intensity

Source	MWh	Renewable MWh	Renewable share
Natural gas boiler, space heating	180.0	0.0	0%
Company vans, diesel	21.1	1.1	5%
Forklift, LPG	7.3	0.0	0%
Purchased grid electricity	200.0	3.2	2%
Total	**408.3**	4.3	1%

Fuels are counted at gross calorific value. The renewable share of electricity follows the contract, as VSME B3 and ESRS E1-5 expect, not the grid average.

Intensity metric	Location-based	Market-based
t CO2e per EUR million revenue	21.03	34.78
kg CO2e per tonne of material processed	133.34	220.53
t CO2e per full-time employee	2.88	4.76
kg CO2e per m2 floor area	34.17	56.51

Intensity figures are what customers usually ask for, and they can be shared without revealing absolute energy use.

Uncertainty and data quality

Every activity row carries a data quality score from 1 (metered or invoiced) to 5 (rough estimate), and every factor carries its own uncertainty. I ran 10,000 Monte Carlo draws, keeping each factor's error the same across all months that use it. The ranges below are 95% intervals. The emissions-weighted data quality score is 1.1.

Total	Central (t)	95% range (t)	±
Scope 1	40.5	39.5 to 41.6	3%
Scope 2, location-based	68.8	65.3 to 72.2	5%
Scope 2, market-based	140.3	133.2 to 147.4	5%
Total, location-based	109.3	105.7 to 112.9	3%
Total, market-based	180.8	173.7 to 188.0	4%

Score	Meaning	Activity uncertainty
1	Metered or invoiced, complete year	±2%
2	Primary data, small gaps estimated	±5%
3	Estimated from partial primary data	±15%
4	Proxy, benchmark or engineering estimate	±30%
5	Rough estimate	±50%

Method

The inventory follows the GHG Protocol Corporate Standard and the Scope 2 Guidance. Every figure is activity data multiplied by an emission factor. I calculate each gas separately (CO_2, CH_4, N_2O and the HFCs in refrigerants), then convert to CO_2e with IPCC AR6 (2021), 100-year GWPs, fossil CH4.

Fuel factors come from the UK Government (DESNZ) 2025 conversion factors, the best-documented free set with a per-gas split. DESNZ publishes them in CO_2e using AR5 values, so I divide back to kg of each gas before applying AR6. Gas is billed in Germany in kWh on a gross calorific value basis (Brennwert), which matches the factor used.

Scope 2 is reported twice, as the Scope 2 Guidance requires. Location-based uses the Umweltbundesamt (UBA) factor for German electricity in 2025 (344 g CO_2/kWh). Market-based follows the guidance's hierarchy: kWh covered by cancelled guarantees of origin count at zero, the supplier's own disclosed factor (Stromkennzeichnung under §42 EnWG) comes next, and the residual mix for Germany (701 g CO_2/kWh, AIB 2025) covers the rest. Both electricity factors are CO_2 only and exclude upstream emissions.

Refrigerant leaks use the GHG Protocol screening method (nameplate charge times an annual leak rate) where there are no service records, and a simplified mass balance (refrigerant topped up during the year) where there are.

Biogenic CO_2 from the biodiesel share in diesel is reported separately, outside the scopes, as the standard asks.

Boundary

The boundary is the Bonn, Germany site under operational control. It covers Scope 1 (fuel burned on site and in company vehicles, plus refrigerant leaks) and Scope 2 (purchased electricity). The site buys no heat or steam.

Scope 3 is outside this boundary. For a converting business it is almost certainly the largest part of the footprint: the filter media that arrive to be cut carry the emissions of their own manufacture. Inbound and outbound freight, waste, business travel and commuting are also Scope 3. I left all of them out on purpose, because Scope 1 and 2 are what the company controls directly and can measure from its own bills.

2025 is the base year. I will recalculate it if a structural change (an acquisition, a new site, outsourcing), a change in method or factors, or a discovered error shifts base-year emissions by 5% or more.

To put a rough number on it: 900 t of incoming media at the DESNZ factor for primary paper production gives about **1,211 t CO_2e**, roughly 7 times Scope 1 and 2 combined (market-based). Paper is only a proxy for specialty filter media, and the real factor could be well above or below it. The order of magnitude is the point: a full footprint would be mostly Scope 3.

Recommendations

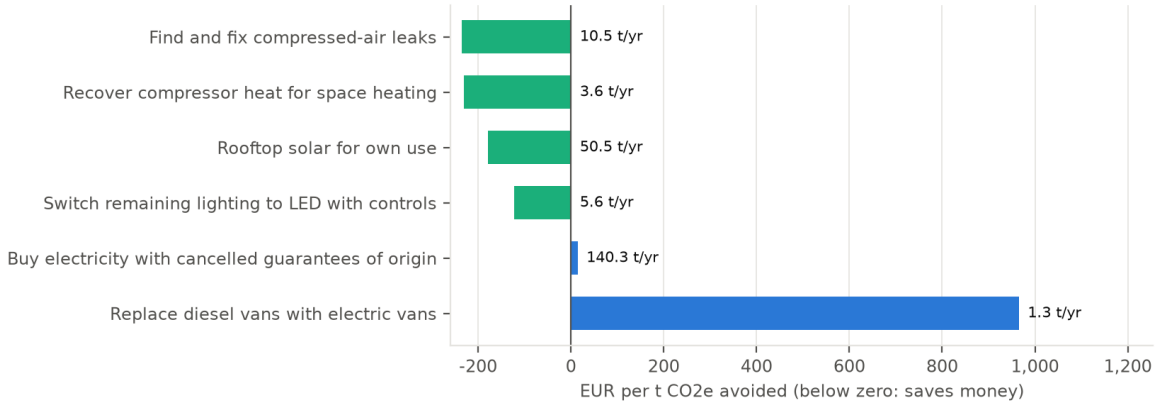
Start with the measures that pay for themselves. On these figures that means: Find and fix compressed-air leaks; Recover compressor heat for space heating; Rooftop solar for own use; Switch remaining lighting to LED with controls. Each costs less over its life than the energy it saves.

Switching to a tariff backed by guarantees of origin would cut market-based emissions by about 140 t a year for roughly EUR 2,000. It is the cheapest way to move the number customers see, but it changes nothing physically: the location-based figure stays put. Do it alongside efficiency, not instead of it. Prefer German or nearby GOs bought from new plants if the claim needs to hold up.

Of the physical measures, 'Rooftop solar for own use' gives the biggest real cut. It saves about 25 t a year location-based (51 t market-based).

One result runs against intuition. The option 'Replace the gas boiler with a heat pump' lowers location-based emissions by 14 t but raises market-based emissions by 5 t, because the extra electricity is counted at the residual mix. Pair electrification with a cleaner electricity contract, or the number customers see goes up.

The national CO_2 price (BEHG, EUR 55/t in 2025) already adds about EUR 2,172 a year to the gas and fuel bills. The corridor is EUR 55 to 65 in 2026, and fuels move into EU ETS2 from 2028 with a market price, so this exposure is likely to grow.



Option	t/yr (LB)	t/yr (MB)	Capex EUR	EUR/t	Payback
Find and fix compressed-air leaks	5.2	10.5	3,000	-235	0.8 yr
Recover compressor heat for space heating	3.6	3.6	9,000	-231	5.1 yr
Rooftop solar for own use Exported surplus is not credited.	24.8	50.5	90,000	-179	5.6 yr
Switch remaining lighting to LED with controls	2.8	5.6	12,000	-122	6.2 yr

Option	t/yr (LB)	t/yr (MB)	Capex EUR	EUR/t	Payback
Buy electricity with cancelled guarantees of origin Changes the market-based figure only. The grid emits the same.	0.0	140.3	0	14	none
Replace diesel vans with electric vans Assumes depot charging on the site meter.	3.3	1.3	20,000	965	10.1 yr
Replace the gas boiler with a heat pump Adds electricity demand.	14.4	-4.9	70,000	n/a	21.6 yr

EUR per tonne is the annualised cost (capex over the asset life at the configured discount rate, plus running costs, minus energy savings) divided by the market-based saving. Below zero means the measure saves money. Options overlap and do not add up.

Where these numbers go

Customers reporting under CSRD will ask suppliers for these figures, usually in the format of the voluntary SME standard (VSME). The table shows where each one belongs.

Framework	Disclosure	Value
GHG Protocol Corporate Standard	Scope 1, gross	40.5 t CO2e
GHG Protocol Scope 2 Guidance	Scope 2, location-based and market-based (dual reporting)	68.8 t / 140.3 t CO2e
GHG Protocol Corporate Standard	Biogenic CO2, reported outside the scopes	0.28 t CO2
VSME (EFRAG) B3	Total energy consumption, renewable and non-renewable	408.3 MWh, of which 4.3 MWh renewable
VSME (EFRAG) B3	Gross Scope 1 and Scope 2 emissions	40.5 t; 68.8 t (LB), 140.3 t (MB)
VSME (EFRAG) B3	GHG intensity per turnover	21.0 (LB) / 34.8 (MB) t CO2e per EUR million
ESRS E1-5	Energy consumption and mix	408.3 MWh total, 1% renewable
ESRS E1-6	Gross Scope 1; Scope 2 location- and market-based; total	40.5; 68.8 / 140.3; 109.3 / 180.8 t CO2e
ESRS E1-6	GHG intensity per net revenue	21.0 (LB) / 34.8 (MB) t CO2e per EUR million
ISO 14064-1:2018	Category 1, direct emissions and removals	40.5 t CO2e
ISO 14064-1:2018	Category 2, indirect emissions from imported energy	68.8 t CO2e (location-based)

Limitations

This edition uses realistic figures for a site of this type, not HTWest's meter readings or invoices. The structure, factors and method are final. The numbers are placeholders until real bills go into the input workbook.

It covers Scope 1 and 2 only, so it is a partial footprint by design. The Scope 3 figure above is a rough screen, not an inventory.

Fuel factors are UK averages for 2025. German diesel and natural gas are close but not identical, and the UBA 2025 grid factor is an estimate that may be revised.

Neither electricity factor includes CH_4, N_2O or upstream emissions. That is normal for Scope 2 in Germany, but it understates the full effect a little.

Where there are no service records, refrigerant leak rates are defaults. Real leak rates vary a lot from one machine to the next.

The abatement costs and savings use typical parameters, not quotes or measurements. The options overlap, so their savings do not add up.

Real HTWest figures only go into a public repository with the company's agreement. Otherwise I would publish intensity figures or an index instead of absolute totals.

Sources

Association of Issuing Bodies (AIB). *AIB European Residual Mixes 2025, final results*. AIB Residual Mixes 2025, Table 2, Germany: 701.47 g CO2/kWh, 1.60% renewable. https://www.aib-net.org/facts/european-residual-mix/2025

Department for Energy Security and Net Zero (DESNZ). *UK Government GHG Conversion Factors for Company Reporting 2025*. Fuels > Liquid fuels > Diesel (average biofuel blend), litres, ID 1_101_1011_8; biogenic CO2 from Outside of scopes ID 99_604_1011_8. https://www.gov.uk/government/publications/greenhouse-gas-reporting-conversion-factors-2025

Department for Energy Security and Net Zero (DESNZ). *UK Government GHG Conversion Factors for Company Reporting 2025*. Fuels > Gaseous fuels > LPG, litres, ID 1_100_1003_8. https://www.gov.uk/government/publications/greenhouse-gas-reporting-conversion-factors-2025

Department for Energy Security and Net Zero (DESNZ). *UK Government GHG Conversion Factors for Company Reporting 2025*. Fuels > Gaseous fuels > Natural gas, kWh (Gross CV), ID 1_100_1004_6. https://www.gov.uk/government/publications/greenhouse-gas-reporting-conversion-factors-2025

Department for Energy Security and Net Zero (DESNZ). *UK Government GHG Conversion Factors for Company Reporting 2025*. Material use > Paper > Paper and board: paper, Primary material production, tonnes, ID 19_506_5164_15. https://www.gov.uk/government/publications/greenhouse-gas-reporting-conversion-factors-2025

Umweltbundesamt (UBA). *Umweltbundesamt: CO2-Emissionen pro Kilowattstunde Strom 2025*. UBA, 344 g CO2/kWh for 2025 (estimate; 2024: 353 g, 2023: 379 g). https://www.umweltbundesamt.de/themen/co2-emissionen-pro-kilowattstunde-strom-2025-nur

World Resources Institute and WBCSD. *GHG Protocol Corporate Accounting and Reporting Standard*, revised edition, and *GHG Protocol Scope 2 Guidance* (2015). https://ghgprotocol.org

IPCC. Global warming potentials: IPCC AR6 (2021), 100-year GWPs, fossil CH4.

EFRAG. *Voluntary Sustainability Reporting Standard for non-listed SMEs (VSME)*, December 2024.

Brennstoffemissionshandelsgesetz (BEHG). National CO_2 price for heating and transport fuels.

Generated 2026-10-17 15:22 UTC.